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# Electromagnetic Induction

*A visual masterclass in flux, induced fields, magnetic energy and modern applications*

October 11, 2026

**Abstract**

A changing magnetic field can drive a current through a circuit with no battery at all. We follow that idea from Faraday's experiments to flux, Faraday's and Lenz's laws, inductance, RL circuits and the AC generator.

## 1 Learning goals

- ☐ Relate magnetic flux change to induced emf.
- ☐ Use Lenz's law to determine current direction.
- ☐ Connect Faraday's law to generators and inductors.

> **How to use these notes.**
>
> Preview the diagram first, cover each worked solution and attempt it yourself, then answer the retrieval questions without looking back. Revisit any question you cannot explain aloud.

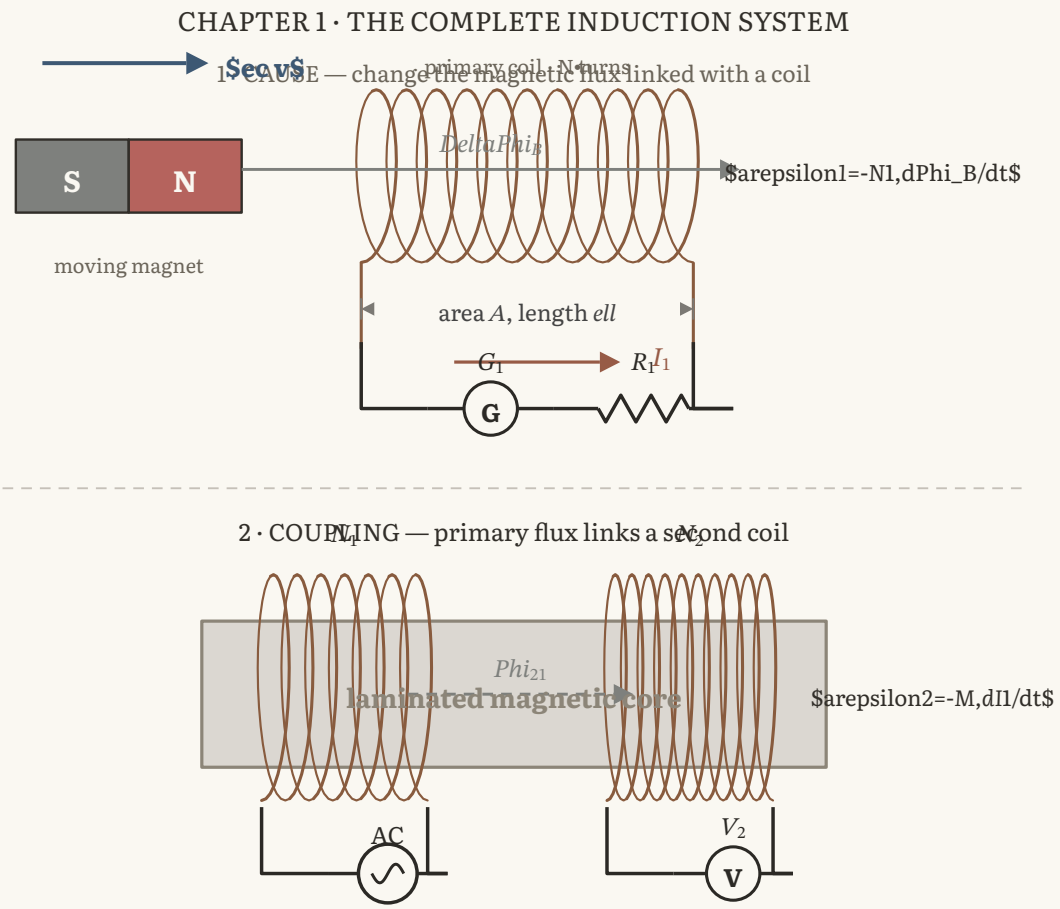



**Figure 1:** Full-system electromagnetic-induction plate. The two experiments share the same chain of reasoning: identify what changes the flux, apply Faraday–Lenz, then follow the induced current and energy.

## 2 The experiments of Faraday and Henry

In 1831 Faraday and Henry found, independently, that a current appears in a closed coil whenever the magnetic field through it changes. A magnet, a coil and a galvanometer are enough to see it.

- A current flows only while the magnet and coil **move relative to each other**; at rest there is none, however strong the magnet.
- Faster motion gives a **larger** current, and reversing the motion **reverses** it.
- Switching the current in a *nearby* coil on or off has the same effect with no motion at all.

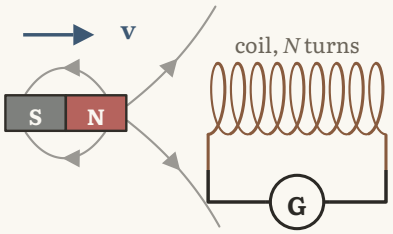



**Figure 2:** Moving the N pole in deflects G; moving it out reverses the deflection.

## 3 Magnetic flux

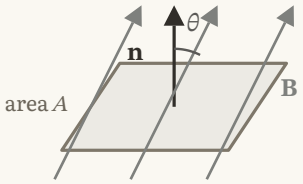



**Figure 3:** Only $B\cos\theta$ threads the area $A$.

Magnetic flux measures how much field threads a surface. It depends on the angle $\theta$ between $\mathbf{B}$ and the surface normal $\mathbf{n}$: it is largest when the surface faces the field and zero when the field skims along it.

**Definition 1** (Magnetic flux).

Measured in webers, $1\ \mathrm{Wb} = 1\ \mathrm{T\,m^2}$. For a curved surface, sum over every element $d\mathbf{A}$:

$$\Phi_B = \vec{B}\cdot\vec{A} = BA\cos\theta, \qquad \Phi_B = \int \vec{B}\cdot d\vec{A} \tag{1}$$

# 4 Faraday’s law of induction

**Faraday’s law.**

The induced emf equals the negative rate of change of the flux linked with the circuit. Since $\Phi_B = BA\cos\theta$, it can come from changing $B$ (transformers), the area $A$ (sliding rods) or the angle $\theta$ (generators). For $N$ turns:

$$\boxed{\varepsilon = -N\,\frac{d\Phi_B}{dt}} \tag{2}$$

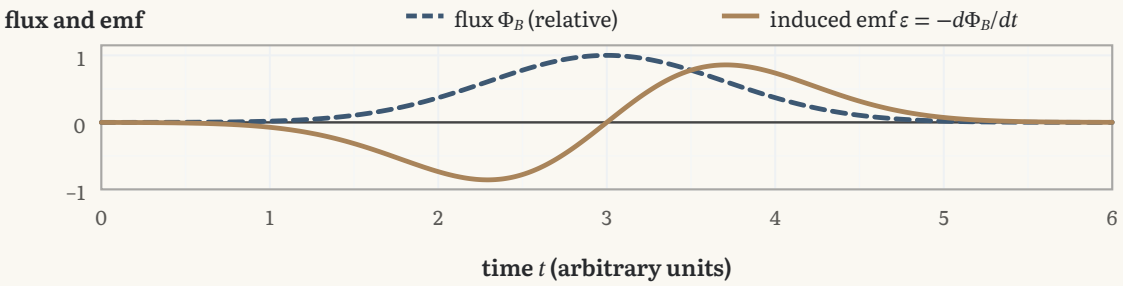



**Figure 4:** A magnet falling through a coil: the flux rises, peaks and falls, so the emf is a pulse that reverses where the flux peaks.

**Example 1** (A coil in a growing field).

A 200-turn coil of area $0.010\ \text{m}^2$ sees $B$ rise from $0.10$ to $0.50\ \text{T}$ in $0.20\ \text{s}$, so $|\varepsilon| = NA\,\Delta B/\Delta t = (200)(0.010)(0.40)/(0.20) = 4.0\ \text{V}$. Through $10\ \Omega$ this drives $I = 0.40\ \text{A}$.

# 5 Lenz’s law: the direction of the current

**Lenz’s law.**

The induced current opposes the change in flux that causes it. This is the minus sign in Faraday’s law, and energy conservation in disguise.

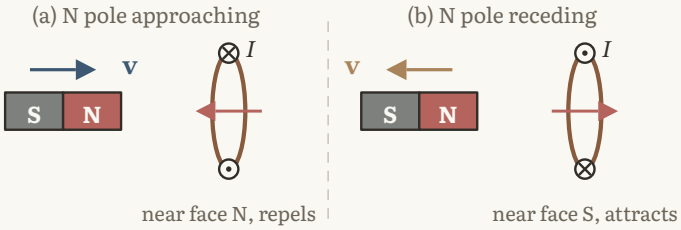



**Figure 5:** The induced field (red) opposes the magnet's while the flux grows (a) and aids it while the flux shrinks (b). Eddy currents in solid metal obey the same rule, which is how magnetic brakes work.

## 6 Motional emf

Flux also changes when the circuit changes shape. A rod of length $\ell$ slides at speed $v$ on rails joined by $R$; the enclosed area $\ell x$ gives $\Phi_B = B\ell x$.

$$\varepsilon = \frac{d\Phi_B}{dt} = B\ell\,\frac{dx}{dt} = B\ell v \tag{3}$$

The force $q\mathbf{v}\times\mathbf{B}$ drives charges up the rod. The current $I = B\ell v/R$ then feels a force $BI\ell$ opposing the motion, so the pushing power $Fv = \varepsilon^2/R$ is exactly the heat produced in $R$.

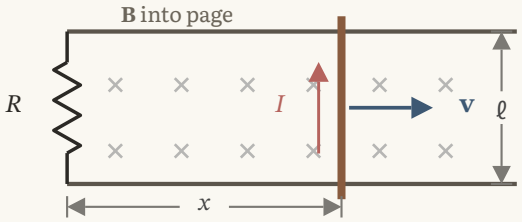



**Figure 6:** The sliding rod sweeps out area, so the flux grows at the rate $B\ell v$.

## 7 Self-inductance

A coil carrying current $I$ links its own flux $N\Phi_B = LI$, where $L$ is the **self-inductance**. The resulting *back emf* resists any change of current, giving the circuit electrical inertia:

$$\varepsilon = -L\,\frac{dI}{dt}, \qquad L_{\text{solenoid}} = \frac{\mu_0 N^2 A}{\ell} \quad \text{(henry, H)} \tag{4}$$

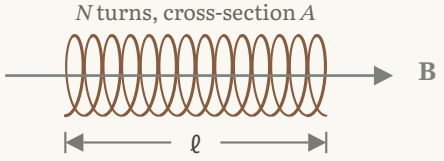



**Figure 7:** Inside a long solenoid, $B = \mu_0 nI$, $n = N/\ell$.

**Example 2** (Inductance of an air-cored coil).

With $N = 500$, $\ell = 25$ cm and $r = 1.0$ cm, $L \approx 0.39$ mH. An iron core multiplies this by $\mu_r$, often over a thousand.

## 8 Mutual inductance

A changing current in one coil (the primary) changes the flux through a neighbour (the secondary) and induces an emf there. This coupling is the principle of every transformer.

The secondary links flux $N_2\Phi_{21} = MI_1$, where $M$ is the **mutual inductance**. For coils on one long former, and in general,

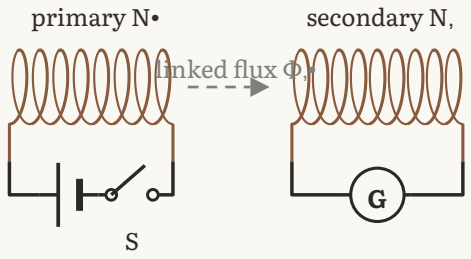



**Figure 8:** Switching $I_1$ on or off makes G kick in the separate secondary.

$$\varepsilon_2 = -M\,\frac{dI_1}{dt}, \qquad M = \frac{\mu_0 N_1 N_2 A}{\ell}, \qquad M = k\sqrt{L_1 L_2}\ \ (0 \le k \le 1) \tag{5}$$

## 9 Energy and the RL circuit

Building up current takes work against the back emf at the rate $LI\,dI/dt$. Integrating gives the stored energy, and its density for any magnetic field:

$$U = \int_0^I LI\,dI = \tfrac{1}{2}LI^2, \qquad u_B = \frac{B^2}{2\mu_0} \tag{6}$$

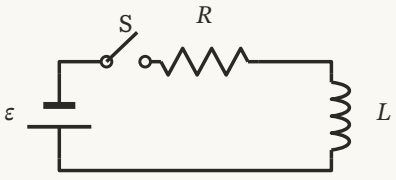



**Figure 9:** A series RL circuit.

The loop rule $\varepsilon = IR + L\,dI/dt$ gives an exponential rise; with the battery removed the current decays:

$$I_{\text{growth}} = \frac{\varepsilon}{R}\left(1 - e^{-t/\tau}\right)$$
$$I_{\text{decay}} = I_0\,e^{-t/\tau}, \quad \tau = \frac{L}{R} \tag{7}$$

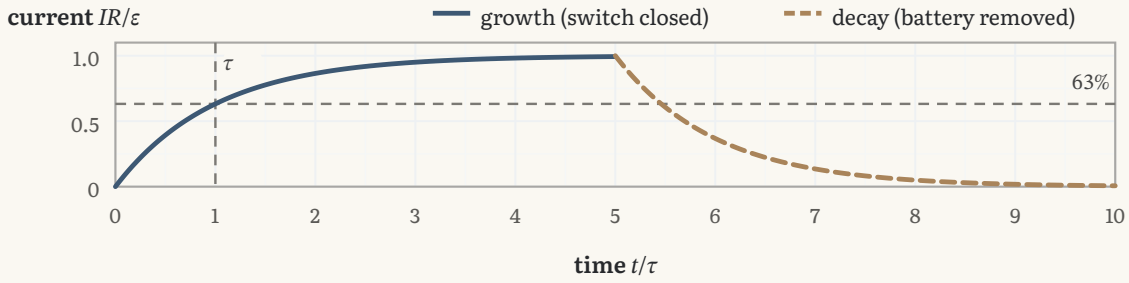



**Figure 10:** After one time constant the current reaches 63% of its final value; after $5\tau$, within 1%.

# 10 The AC generator

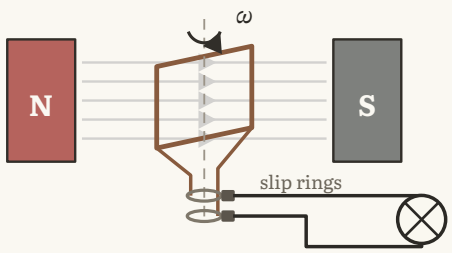



**Figure 11:** A simple AC generator.

A coil of $N$ turns and area $A$ rotating at angular speed $\omega$ in a uniform field has $\theta = \omega t$, so its flux varies as a cosine and the emf as a sine.

$$\Phi_B = BA\cos\omega t$$
$$\varepsilon = -N\frac{d\Phi_B}{dt} = NBA\omega\,\sin\omega t \tag{8}$$

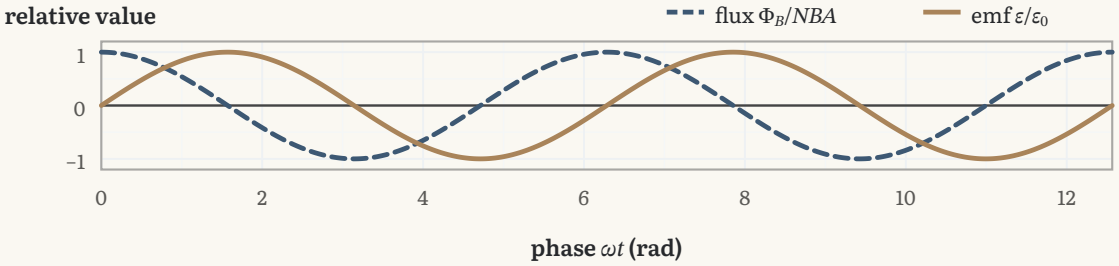



**Figure 12:** The emf peaks when the flux is zero, and vanishes when the flux peaks.

> **Example 3** (Mains frequency).
>
> 100 turns of area 0.050 m² turning at 50 Hz in 0.20 T: $\omega = 314$ rad s$^{-1}$, so $\varepsilon_0 = NBA\omega = 314$ V.

# 11 Teacher viva: questions you must answer aloud

Practise these without looking at the formulas. Begin with the physical idea, state the relevant law, then give the mathematical relation with units and assumptions.

## 11.1 Rapid-fire conceptual questions

1. **Why is no current induced when a magnet is held still inside a coil?** Flux may be large, but $d\Phi_B/dt = 0$.
2. **What does the minus sign in Faraday's law mean?** The induced effect opposes the *change* in linked flux; it expresses Lenz's law and energy conservation.
3. **Can emf exist without current?** Yes. An open circuit can have induced emf, but current requires a closed conducting path.
4. **How can flux change if $B$ is constant?** Change area, orientation, or the fraction of the field intercepted by the loop.

5. **Why does a transformer require alternating current?** A steady primary current gives steady flux, so the secondary has no sustained induced emf.
6. **Why are transformer cores laminated?** Thin insulated sheets interrupt eddy-current loops and reduce heating.
7. **What sets the direction of motional emf?** The magnetic force $q\mathbf{v} \times \mathbf{B}$, combined with Lenz's law for the complete circuit.
8. **Where is energy stored in an inductor?** In its magnetic field, with $U = LI^2/2$ and density $u_B = B^2/(2\mu)$.
9. **Why can current through an ideal inductor not jump instantly?** An instantaneous change would require infinite $dI/dt$ and therefore infinite back emf.
10. **At $t = \tau$ in an RL growth circuit, what has happened?** Current reaches $1 - e^{-1} = 63.2\%$ of its final value.
11. **Flux and emf in a generator are how far out of phase?** One quarter-cycle, or $\pi/2$ radians.
12. **Does Lenz's law oppose motion in every case?** It opposes the flux change; magnetic drag is the mechanical consequence when motion causes that change.

### 11.2 Exam-level numerical checklist

- ☐ I write the flux linkage $N\Phi_B$ before differentiating.
- ☐ I convert centimetres to metres and milliseconds to seconds.
- ☐ I separate emf magnitude from direction.
- ☐ I state whether the circuit is open or closed before finding current.
- ☐ I check dimensions: $\text{Wb/s} = \text{V}$ and $\text{H}/\Omega = \text{s}$.
- ☐ I test limiting cases such as $t = 0$, $t \to \infty$, or $\theta = 90°$.

> **A strong oral answer.**
>
> Use this four-part pattern: **observation → law → equation → consequence**. Example: "The linked flux is decreasing; by Lenz's law the coil reinforces the original field; Faraday's law gives $|\varepsilon| = N|d\Phi/dt|$; therefore the induced current reverses when the motion reverses."

## 12 Summary

**Table 1:** The key results of the chapter.

| Quantity | Relation | SI unit |
|---|---|---|
| Magnetic flux | $\Phi_B = BA\cos\theta$ | weber (Wb) |

| | | |
|---|---|---|
| Faraday's law | $\varepsilon = -N\,d\Phi_B/dt$ | volt (V) |
| Motional emf | $\varepsilon = B\ell v$ | volt (V) |
| Self-inductance | $\varepsilon = -L\,dI/dt,\ L = \mu_0 N^2 A/\ell$ | henry (H) |
| Mutual inductance | $\varepsilon_2 = -M\,dI_1/dt$ | henry (H) |
| Energy in an inductor | $U = LI^2/2$ | joule (J) |
| RL time constant | $\tau = L/R$ | second (s) |
| AC generator | $\varepsilon = NBA\omega \sin \omega t$ | volt (V) |

## 13 Active recall

Close the notes and answer these from memory. A strong answer should include the physical reason, not only a formula.

1. **Question:** Can a strong constant magnetic field induce emf?
2. **Question:** What does the minus sign in Faraday's law mean?
3. **Question:** How can flux change if field strength stays constant?

> **Common exam traps.**
>
> - Flux, not magnetic field alone, controls induced emf.
> - Lenz's law opposes the change in flux, not necessarily the field itself.

### 13.1 Final self-check

- [ ] I can sketch and label the central diagram from memory.
- [ ] I can derive the key result and state its assumptions.
- [ ] I can solve a numerical example with units and a reasonableness check.
- [ ] I can explain the topic in sixty seconds without reading.